

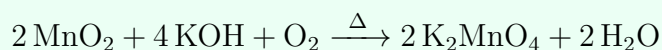
Chapter 4: d- and f-Block Elements

Named Reactions + Tips & Tricks

Formula Sheet

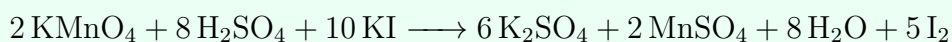
I. Potassium Permanganate (KMnO_4) – sp^3 , tetrahedral

Preparation from Pyrolusite Ore



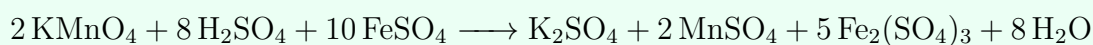
Potassium manganate (green mass) is formed first; oxidised further to permanganate.

Oxidation of Iodide to Iodine (Basic medium)



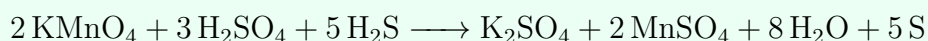
Basic medium; KI is oxidised to I_2 .

Ferrous Salts to Ferric Salts (Acidic)



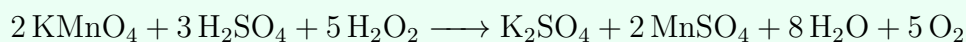
Acidic medium; ferrous sulphate oxidised to ferric sulphate.

Hydrogen Sulphide to Sulphur (Neutral)



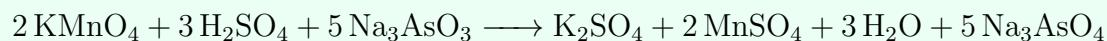
Neutral medium; H_2S oxidised to elemental sulphur.

Hydrogen Peroxide to Water (Acidic)

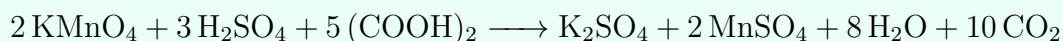


Acidic medium; used to standardise H_2O_2 solutions.

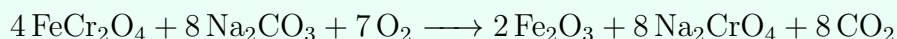
Arsenites to Arsenates



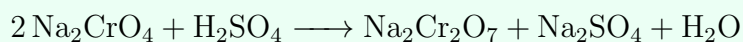
Sodium arsenite oxidised to sodium arsenate.

Oxalic Acid (Oxalates) to Carbon Dioxide

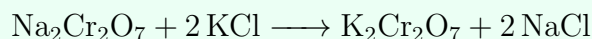
Classic redox titration: KMnO_4 vs. oxalic acid.

II. Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) – Cr in +6 O.S., two tetrahedral CrO_4 units sharing a corner O**Preparation – Chromite Ore to Sodium Chromate**

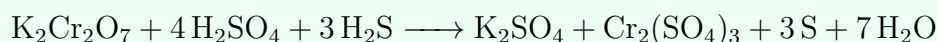
Roasting chromite ore with sodium carbonate and air.

Chromate to Dichromate

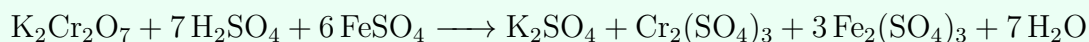
Acidification converts yellow chromate to orange dichromate.

Sodium Dichromate to Potassium Dichromate

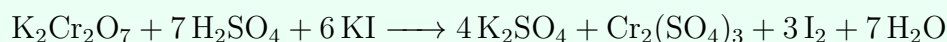
Final step of extraction from chromite ore.

Oxidation: Hydrogen Sulphide to Sulphur

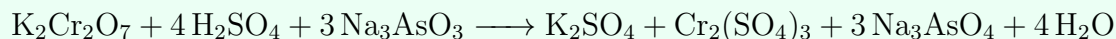
Acidic medium.

Oxidation: Ferrous to Ferric Salts

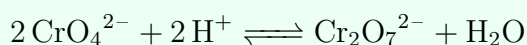
Acidic medium; classic redox titration reaction.

Oxidation: Iodide to Iodine

Acidic medium.

Oxidation: Arsenites to Arsenates

Acidic medium.

Chromate-Dichromate Interconversion (pH dependent)

Yellow chromate ion and orange dichromate ion exist in a pH-dependent equilibrium.

Tips, Tricks & Hacks

- **"Why are enthalpies of atomisation of transition metals high?"** – Strong metallic bonds between atoms of these elements, so a large amount of heat is required to free the atoms.
- **"Why do transition metals form coloured compounds?"** – Incomplete *d*-subshell / unpaired electrons in the *d*-subshell allow *d*-*d* transitions, so they become coloured.
- **"Why do transition metals and their compounds show paramagnetic behaviour?"** – They have unpaired electrons in the $(n - 1)d$ shell.
- **"Why is the lowest oxide of a transition metal basic while the highest is acidic?"** – At low oxidation states, valence electrons are only partly involved in bonding, so the metal can donate electrons and behave as a base; at high oxidation states, all valence electrons are engaged in bonding, so it can accept electrons and behaves as an acid.
- **"Why is copper considered a transition metal?"** – Cu in the +2 oxidation state has configuration $4s^0 3d^9$, which is an incompletely filled *d*-subshell.
- **"Why is chromium a typical hard metal but mercury a liquid?"** – Cr has 5 unpaired *d*-electrons ($3d^5 4s^1$) giving very strong metallic bonds; Hg has fully filled $3d^{10} 4s^0$ orbitals, so metallic bonding is very weak.
- **"Why is scandium colourless but titanium coloured?"** – Sc has no unpaired electrons; Ti has 1 unpaired electron (enabling *d*-*d* transition).
- **"Why do transition metals have high enthalpies of hydration?"** – Small atomic/ionic size and high nuclear charge, both of which increase left to right along the series.
- **"Why do transition metals show variable oxidation states?"** – The energies of $(n - 1)d$ and *ns* orbitals are very close, so electrons from both can take part in bonding.
- **"Why do transition elements show irregularities in electronic configuration?"** – Very small energy difference between $(n - 1)d$ and *ns* subshells means the incoming electron may enter either subshell (e.g., Cr: $3d^5 4s^1$, Cu: $3d^{10} 4s^1$ – extra stability of half-filled/fully filled *d*).
- **"Why does paramagnetic character in the 3d series increase up to Mn and then decrease?"** – Unpaired electrons increase from Sc to Mn; after Mn, pairing of electrons begins and the number of unpaired electrons decreases.
- **"Why are zinc salts white but Cu^{2+} salts blue, though both are *d*-block?"** – Zn has a fully filled *d*-subshell (no *d*-*d* transition possible); Cu^{2+} has a partially filled *d*-subshell.
- **Oxidation state trend:** minimum is +2, maximum is +8 (highest with O and F as ligands); bonds with lower O.S. are ionic, with higher O.S. are covalent.
- **Lanthanoid contraction** causes near-identical radii in 4d/5d congeners, difficulty separating lanthanoids, and decreasing basicity of oxides/hydroxides down the series.